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PROCESSING METHOD AND COMPUTER-READABLE STORAGE MEDIUM

Abstract

A processing method is a method of processing a workpiece including a base material and a device layer formed on the base material. The workpiece has a plurality of planned division lines. The device layer is formed by stacking a plurality of layers. The processing method includes: denaturing a specific layer constituting the device layer by applying a laser beam along the planned division lines of the workpiece via the base material; and after denaturing the specific layer, by applying a laser beam along the planned division lines, forming a modified layer along the planned division lines in the base material and forming a crack extending from the modified layer for dividing the device layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] The present application claims priority to and incorporates by reference the entire contents of Japanese Patent Application No. 2024-036210 filed in Japan on Mar. 8, 2024.

TECHNICAL FIELD

[0002] The present disclosure relates to a processing method of a workpiece having a base material and a device layer, and a computer-readable storage medium storing a laser beam processing program.

BACKGROUND

[0003] For example, a processing method has been proposed, which uses a laser beam to divide a semiconductor wafer in which a device layer constituting devices is formed on a semiconductor base material (see, for example, JP 2005-86161 A).

[0004] In the processing method disclosed in JP 2005-86161 A, a modified layer is formed inside the semiconductor base material by applying a laser beam to the semiconductor wafer along streets while condensing the laser beam inside the semiconductor wafer, and cracks are formed to extend from the modified layer toward the surface of the semiconductor wafer.

[0005] However, in the processing method disclosed in JP 2005-86161 A, the cracks may not extend straight from the modified layer but extend obliquely, or may extend bifurcating from the modified layer, depending on the thickness, material, and the like of the device layer. In the case of such a division defect, chips formed by dividing the semiconductor wafer differ in the outer diameter size, and thus improvement has been desired.

SUMMARY

[0006] A processing method according to the present disclosure is of processing a workpiece including a base material and a device layer formed on the base material. The workpiece has a plurality of planned division lines. The device layer is formed by stacking a plurality of layers. The processing method includes: denaturing a specific layer constituting the device layer by applying a laser beam along the planned division lines of the workpiece via the base material; and after denaturing the specific layer, by applying a laser beam along the planned division lines, forming a modified layer along the planned division lines in the base material and forming a crack extending from the modified layer for dividing the device layer.

[0007] A non-transitory computer readable storage medium according to the present disclosure stores a laser beam processing program for processing a workpiece including a base material and a device layer formed on the base material. The workpiece has a plurality of planned division lines. The device layer is formed by stacking a plurality of layers. The laser beam processing program causes a computer to execute: denaturing a specific layer constituting the device layer by applying a laser beam along the planned division lines of the workpiece via the base material; and after denaturing the specific layer, by applying a laser beam along the planned division lines, forming a modified layer in the base material along the planned division lines and forming a crack extending from the modified layer for dividing the device layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a perspective view illustrating a configuration example of a workpiece to be processed by a processing method according to an embodiment;

[0009] FIG. **2** is a cross-sectional view schematically illustrating main parts of the workpiece illustrated in FIG. **1**;

[0010] FIG. **3** is a flowchart illustrating the flow of the processing method according to the embodiment;

[0011] FIG. **4** is a perspective view illustrating a configuration example of a laser beam processing machine that performs a specific layer denaturation step and a modification step of the processing method illustrated in FIG. **3**;

[0012] FIG. **5** is a diagram illustrating the configuration of a laser beam application unit of the laser beam processing machine illustrated in FIG. **4**;

[0013] FIG. **6** is a cross-sectional view schematically illustrating the main parts of the workpiece with a tape attached to its surface in the specific layer denaturation step of the processing method illustrated in FIG. **3**;

[0014] FIG. **7** is a cross-sectional view of the workpiece schematically illustrating a state in which a laser beam is applied to the workpiece in the specific layer denaturation step of the processing method illustrated in FIG. **3**;

[0015] FIG. **8** is a plan view of the workpiece schematically illustrating pulsed laser beam spots where the laser beam is applied to a specific layer illustrated in FIG. **7**;

[0016] FIG. **9** is a cross-sectional view schematically illustrating the main parts of the workpiece during the modification step of the processing method illustrated in FIG. **3**;

[0017] FIG. **10** is a cross-sectional view schematically illustrating the main parts of the workpiece after the modification step of the processing method illustrated in FIG. **3**;

[0018] FIG. **11** is a perspective view schematically illustrating a grinding step of the processing method illustrated in FIG. **3**; and

[0019] FIG. **12** is a cross-sectional view schematically illustrating the main parts of the workpiece after the grinding step of the processing method illustrated in FIG. **3**.

DETAILED DESCRIPTION

[0020] An embodiment of the present disclosure will be described in detail with reference to the drawings. The present invention is not limited by the contents described in the following embodiment. In addition, constituent elements described below include those that can be easily assumed by those skilled in the art and those that are substantially the same. Furthermore, configurations described below can be appropriately combined. In addition, various omissions, substitutions, or changes in the configurations can be made without departing from the gist of the present invention.

[0021] A processing method according to an embodiment of the present invention will be described with reference to the drawings. FIG. **1** is a perspective view illustrating a configuration example of a workpiece to be processed by the processing method according to the embodiment. FIG. **2** is a cross-sectional view schematically illustrating main parts of the workpiece illustrated in FIG. **1**. FIG. **3** is a flowchart illustrating the flow of the processing method according to the embodiment.

Workpiece

[0022] The processing method according to the embodiment is a processing method of a workpiece **200** illustrated in FIG. **1**. As illustrated in FIG. **1**, the workpiece **200** to be processed by the processing method according to the embodiment is a wafer, for example, a disk-shaped semiconductor wafer using silicon, sapphire, gallium, SiC, or the like as a base material **201**, or an optical device wafer. As illustrated in FIG. **1**, the workpiece **200** includes devices **204** formed at respective areas on a front surface **202**, which are sectioned in a lattice pattern by planned division lines **203** intersecting each other. The devices **204** are, for example, integrated circuits (IC), large

scale integration (LSI) integrated circuits, or memories (semiconductor storage devices). In this manner, the plurality of planned division lines **203** are set on the workpiece **200**.
[0023] As illustrated in FIGS. **1** and **2**, the workpiece **200** includes the base material **201** and a device layer **205** formed on the base material **201**. The device layer **205** includes a specific layer **206** formed on the base material **201** and a functional layer **207** formed on the specific layer **206**. In the embodiment, the specific layer **206** is a metal layer made of a metal that is a ductile material. The functional layer **207** includes: low dielectric constant insulator films (hereinafter, referred to as low-k films) made of a film of an inorganic material such as SiOF or BSG (SiOB), a film of an organic material such as a polyimide-based or parylene-based polymer, or carbon-containing silicon oxide (SiOCH); and a circuit layer including a conductive metal pattern and a metal film.
[0024] The low-k films, together with the circuit layer, are stacked to form the devices **204**. The circuit layer constitutes the circuits of the devices **204**. Thus, the devices **204** are constituted by: the low-k films stacked on each other of the functional layer **207** formed on the specific layer **206** on the base material **201**; and the circuit layer situated between the low-k films. At areas of the planned division lines **203**, the device layer **205** is constituted by: the specific layer **206**; and the low-k films stacked on the specific layer **206** except for an area of a test elementary group (TEG). As described above, the device layer **205** includes the specific layer **206** at least at areas overlapping the planned division lines **203**, and is formed by stacking a plurality of layers.
[0025] In the present invention, the material of the base material **201** of the workpiece **200** and the type of the devices **204** are not limited to those described in the embodiment. In the present invention, the specific layer **206** is not limited to a metal layer made of a metal, and may be a resin layer made of a resin that is a ductile material. If the specific layer **206** is a resin layer, it is desirable that the specific layer **206** be an insulator layer made of a polymer-based insulator which is a resin. In addition, the specific layer **206** is not limited to a layer interposed between the base material **201** and the functional layer **207**, and may be a layer in the functional layer **207**.
Processing Method
[0026] Next, the processing method according to the embodiment will be described. As illustrated in FIG. **3**, the processing method according to the embodiment includes a specific layer denaturation step **1001**, a modification step **1002**, and a grinding step **1003**. The specific layer denaturation step **1001** and the modification step **1002** are performed by a laser beam processing machine **1** illustrated in FIG. **4**.
Laser Beam Processing Machine
[0027] Next, the laser beam processing machine **1** illustrated in FIG. **4** will be described. FIG. **4** is a perspective view illustrating a configuration example of the laser beam processing machine that performs the specific layer denaturation step and the modification step of the processing method illustrated in FIG. **3**. FIG. **5** is a diagram illustrating the configuration of a laser beam application unit of the laser beam processing machine illustrated in FIG. **4**. As illustrated in FIG. **4**, the laser beam processing machine **1** includes a holding unit **10**, a moving unit **30**, a laser beam application unit **20**, an image pick-up unit (not illustrated), a cleaning unit **40**, a conveyance unit **50**, and a control unit **100**.
[0028] The holding unit **10** includes a holding surface **11** that is disk-shaped. The holding surface **11** is flat in the horizontal direction in which the workpiece **200** is held, and is formed of porous ceramic or the like. In addition, the holding unit **10** is provided to be movable by the moving unit **30** over a processing area under the laser beam application unit **20**, and a loading/unloading area which is separated from the area under the laser beam application unit **20** and where the workpiece **200** is loaded and unloaded.
[0029] The holding unit **10** is connected to a vacuum suction source (not illustrated), and sucks and holds the workpiece **200** placed on the holding surface **11** by being sucked by the vacuum suction source. In the embodiment, the holding unit **10** sucks and holds the front surface **202** side of the workpiece **200** via a tape **209** attached to the front surface **202** of the workpiece **200**.

[0030] The moving unit **30** relatively moves the holding unit **10** and the laser beam application unit **20**. The moving unit **30** includes at least: a Y-axis moving unit **31** which is an indexing-feed unit that moves the holding unit **10** in a Y-axis direction parallel to the horizontal direction; an X-axis moving unit **32** which is a processing-feed unit that moves the holding unit **10** in an X-axis direction parallel to the horizontal direction and orthogonal to the Y-axis direction; and a rotational moving unit **33** that rotates the holding unit **10** about a central axis parallel to a Z-axis direction parallel to the vertical direction.

[0031] The Y-axis moving unit **31** is placed on a machine body **2**, and moves the holding unit **10** in the Y-axis direction by moving a moving plate **3** on which the X-axis moving unit **32** is placed in the Y-axis direction. The X-axis moving unit **32** is placed on the moving plate **3**, and moves the holding unit **10** in the X-axis direction by moving a second moving plate **4** on which the rotational moving unit **33** is placed in the X-axis direction. The rotational moving unit **33** is placed on the second moving plate **4** and supports the holding unit **10**, thereby rotating the holding unit **10** about the central axis.

[0032] The Y-axis moving unit **31** moves the moving plate **3** in the Y-axis direction, together with the X-axis moving unit **32**, the second moving plate **4**, the rotational moving unit **33**, and the holding unit **10**. The X-axis moving unit **32** moves the second moving plate **4** in the X-axis direction, together with the rotational moving unit **33** and the holding unit **10**.

[0033] The Y-axis moving unit **31** and the X-axis moving unit **32** include a known ball screw provided to be rotatable about a central axis, a known motor that rotates the ball screw about the central axis, and a known guide rail that supports the moving plates **3** and **4** such that they are movable in the X-axis direction or the Y-axis direction. The rotational moving unit **33** includes a known motor or the like that rotates the holding unit **10** about the central axis.

[0034] As illustrated in FIG. **4**, a portion of the laser beam application unit **20** is provided at the distal end of a support column **6** whose proximal end portion is supported by an upright wall **5** erected from an end of the machine body **2** in the Y-axis direction. The laser beam application unit **20** applies a laser beam **21** (illustrated in FIG. **5**) to the workpiece **200** held by the holding unit **10** to perform laser beam processing.

[0035] As illustrated in FIG. **5**, the laser beam application unit **20** includes an oscillator **22**, an attenuator **23** which is an output adjustment unit, and a condensing unit **24**. The oscillator **22** is a device that generates and oscillates the pulsed laser beam **21** with a wavelength capable of passing through the workpiece **200**.

[0036] The attenuator **23** adjusts the output of the laser beam **21** oscillated by the oscillator **22**. The condensing unit **24** includes a condensing lens **241** that condenses the laser beam **21** generated and oscillated by the oscillator **22** on the workpiece **200** held by the holding unit **10**, and a condensing lens lifting unit **242** that moves the condensing lens **241** up and down in the Z-axis direction to move a focal point **211** of the laser beam **21** up and down in the Z-axis direction.

[0037] The image pick-up unit includes an image pick-up device that takes an image of an area to be divided of the workpiece **200** held by the holding unit **10** before laser beam processing. The image pick-up device is, for example, a charge-coupled device (CCD) image pick-up device or a complementary MOS (CMOS) image pick-up device. The image pick-up unit takes an image of the workpiece **200** held by the holding unit **10**, obtains an image for performing alignment for aligning the workpiece **200** and the laser beam application unit **20**, and outputs the obtained image to the control unit **100**.

[0038] The cleaning unit **40** cleans the workpiece **200** after laser beam processing. The cleaning unit **40** includes a spinner table **41** with a holding surface that is disk-shaped, and a cleaning nozzle **42**. The holding surface is flat in the horizontal direction in which the workpiece **200** is held, and is formed of porous ceramic or the like. The spinner table **41** is rotated about a central axis parallel to the Z-axis direction by a rotary drive source (not illustrated).

[0039] The spinner table **41** sucks and holds the workpiece **200** placed on the holding surface with

the holding surface connected to a vacuum suction source (not illustrated) and sucked by the vacuum suction source. In the embodiment, the spinner table **41** sucks and holds the front surface **202** side of the workpiece **200** via a tape **209**.
[0040] The cleaning nozzle **42** supplies cleaning water (in the embodiment, pure water) to the front surface **202** of the workpiece **200** held by the spinner table **41**, and cleans a back surface **208** on the back side of the front surface **202** of the workpiece **200**.
[0041] The conveyance unit **50** conveys the workpiece **200** between the holding unit **10** and the cleaning unit **40**. The conveyance unit **50** includes conveyance arms **51** that convey the workpiece **200** between the holding unit **10** and the cleaning unit **40**.
[0042] The control unit **100** controls each constituent element of the laser beam processing machine **1** to cause the laser beam processing machine **1** to perform a processing operation on the workpiece **200**. The control unit **100** is a computer including an arithmetic processing device including a microprocessor such as a central processing unit (CPU), a storage device including a memory such as a read only memory (ROM) or a random access memory (RAM), and an input/output interface device. The arithmetic processing device of the control unit **100** performs arithmetic processing according to a computer program stored in the storage device, and outputs a control signal for controlling the laser beam processing machine **1** to each constituent element of the laser beam processing machine **1** via the input/output interface device.
[0043] The control unit **100** is connected to a display unit (not illustrated) including a liquid crystal display device or the like that displays a state, an image, or the like of the processing operation, and an input unit (not illustrated) used when an operator registers processing content information or the like. The input unit includes at least one of a touch panel provided on the display unit and an external input device such as a keyboard.
[0044] As illustrated in FIG. **1**, the control unit **100** includes a processing controller **101** and a storage **102**. The processing controller **101** controls each constituent element of the laser beam processing machine **1** to cause the laser beam processing machine **1** to perform the processing operation on the workpiece **200**.
[0045] The storage **102** stores a laser beam processing program **103**. The laser beam processing program **103** is a computer program that causes the control unit **100**, which is a computer, that is, the laser beam processing machine **1** to perform the specific layer denaturation step **1001** and the modification step **1002** to perform laser beam processing on the workpiece **200**.
[0046] The function of the storage **102** is implemented by the storage device described above. The function of the processing controller **101** is implemented by the arithmetic processing device described above performing arithmetic processing according to a computer program stored in the storage device.
Specific Layer Denaturation Step
[0047] Next, the specific layer denaturation step **1001** will be described. FIG. **6** is a cross-sectional view schematically illustrating the main parts of the workpiece with the tape attached to its surface in the specific layer denaturation step of the processing method illustrated in FIG. **3**. FIG. **7** is a cross-sectional view of the workpiece schematically illustrating a state in which a laser beam is applied to the workpiece in the specific layer denaturation step of the processing method illustrated in FIG. **3**. FIG. **8** is a plan view of the workpiece schematically illustrating pulsed laser beam spots where the laser beam is applied to the specific layer illustrated in FIG. **7**.
[0048] The specific layer denaturation step **1001** is a step of denaturing the specific layer **206** by applying the laser beam **21** to the specific layer **206** along the planned division lines **203** of the workpiece **200** via the base material **201**. First, in the specific layer denaturation step **1001** of the embodiment, the disk-shaped tape **209** equal in diameter to the workpiece **200** is attached to the front surface **202** of the workpiece **200** as illustrated in FIG. **6**. In the specific layer denaturation step **1001** of the embodiment, the laser beam processing machine **1** starts the processing operation, that is, the execution of the laser beam processing program **103**, and starts the specific layer

denaturation step **1001**, when an operator or the like registers processing conditions in the control unit **100**, the front surface **202** side of the workpiece **200** is placed on the holding surface **11** of the holding unit **10** via the tape **209**, and the control unit **100** receives an instruction to start the processing operation from the operator or the like.

[0049] The processing conditions include the repetition frequency of the pulsed laser beam **21**, the output of the laser beam **21**, the relative movement rates (referred to as processing-feed rates) in the X-axis direction of the laser beam application unit **20** and the holding unit **10**, the position in the thickness direction of the workpiece **200** of the focal point **211** (illustrated in FIG. **5** and the like) of the laser beam **21**, and the like, in the specific layer denaturation step **1001** and the modification step **1002**.

[0050] In the specific layer denaturation step **1001** of the embodiment, the processing controller **101** of the control unit **100** of the laser beam processing machine **1** according to the embodiment sucks and holds the front surface **202** side of the workpiece **200** on the holding surface **11** of the holding unit **10** via the tape **209**. In the specific layer denaturation step **1001** of the embodiment, the processing controller **101** of the control unit **100** of the laser beam processing machine **1** controls the moving unit **30** to move the holding unit **10** toward the processing area, takes an image of the workpiece **200** by the image pick-up unit, and performs alignment based on the image taken by the image pick-up unit. In the present invention, the image pick-up unit may be, for example, an infrared camera, and alignment may be performed based on an image taken from the back surface **208** side of the workpiece **200** by the infrared camera, which is the image pick-up unit. At least part of the holding surface **11** of the holding unit **10** may be made of a transparent member, and the alignment may be performed by taking an image from the front surface **202** side of the workpiece **200** via the transparent member.

[0051] In the specific layer denaturation step **1001** of the embodiment, as illustrated in FIG. **7**, the processing controller **101** of the control unit **100** of the laser beam processing machine **1** sets the focal point **211** of the laser beam **21** inside the tape **209**, and applies the laser beam **21** to the center in the width direction of the planned division lines **203** of the workpiece **200** from the back surface **208** side while controlling the laser beam application unit **20** and the moving unit **30** to relatively move the condensing unit **24** of the laser beam application unit **20** and the holding unit **10** in the X-axis direction along the planned division lines **203**. As described above, in the specific layer denaturation step **1001** of the embodiment, the laser beam processing machine **1** applies the laser beam **21** while positioning the focal point **211** of the laser beam **21** at a position different from the specific layer **206**.

[0052] In this way, in the specific layer denaturation step **1001** of the embodiment, the laser beam processing machine **1** performs so-called defocusing to apply the laser beam **21** to the specific layer **206**, and as illustrated in FIG. **8**, expands the diameter of pulsed laser beam spots **212** in the specific layer **206** to be larger than that of the focal point **211**. Further, in the specific layer denaturation step **1001** of the embodiment, the laser beam processing machine **1** expands the diameter of the pulsed laser beam spots **212** in the specific layer **206** to be larger than that of the focal point **211**, and as illustrated in FIG. **8**, applies the laser beam **21** to an area of the specific layer **206** having a predetermined width **213** in a direction **203-2** orthogonal to an extending direction **203-1** of the planned division line **203** to which the laser beam **21** is applied.

[0053] In the specific layer denaturation step **1001** of the embodiment, in the laser beam processing machine **1**, a processing-feed rate and a repetition frequency is set such that the pulsed laser beam spots **212** in the specific layer **206** partially overlap each other. As described above, in the specific layer denaturation step **1001** of the embodiment, the pulsed laser beam **21** is applied along the planned division lines **203**, such that the pulsed laser beam spot **212** where one pulsed laser beam **21** is applied to the specific layer **206** and the pulsed laser beam spot **212** where the next pulsed laser beam **21** is applied to the specific layer **206** partially overlap each other. In addition, in the specific layer denaturation step **1001** of the embodiment, in the laser beam processing machine **1**, the output of the laser beam **21** is set to a value that does not exceed a processing threshold value of the base material **201** (in the embodiment, a value of approximately 1/10 of the output in the modification step **1002**) in a state where the focal point **211** is set at the above-described position. Thus, the laser beam processing machine **1** does not process the base material **201**.

[0054] In the specific layer denaturation step **1001** of the embodiment, the condensing lens **241** performs so-called defocusing of the laser beam **21** for the specific layer **206**. Alternatively, in the present invention, a beam expander or a spatial light modulator may expands the diameter of the pulsed laser beam spots **212** of the laser beam **21** generated by the oscillator **22** for the specific layer **206**, instead of the defocusing. In the specific layer denaturation step **1001** of the present invention, the pulsed laser beam spots **212** on the specific layer **206** may not partially overlap each other, and a laser beam of a continuous wave (CW) may be applied instead of the pulsed laser beam **21**.

[0055] In the specific layer denaturation step **1001** of the embodiment, the laser beam processing machine **1** applies the defocused laser beam **21** to the specific layer **206** of all the planned division lines **203**, heats and denatures the specific layer **206** along the planned division lines **203**, and forms a denatured portion **210** in the specific layer **206**. The denatured portion **210** has mechanical strength lower than that of the other specific layer **206**, and is more easily divided than the other specific layer **206** by an external force. In the specific layer denaturation step **1001** of the embodiment, the laser beam processing machine **1** forms the denatured portion **210** in the specific layer **206** along all the planned division lines **203**.

Modification Step

[0056] Next, the modification step **1002** will be described. FIG. **9** is a cross-sectional view schematically illustrating the main parts of the workpiece during the modification step of the processing method illustrated in FIG. **3**. FIG. **10** is a cross-sectional view schematically illustrating the main parts of the workpiece after the modification step of the processing method illustrated in FIG. **3**. The modification step **1002** is a step of forming a modified layer **220** along the planned division lines **203** in the base material **201** and forming cracks **221** extending from the modified layer **220** for dividing the device layer **205**, by applying the laser beam **21** along the planned division lines **203** after performing the specific layer denaturation step **1001**.

[0057] The modified layer **220** means an area where density, a refractive index, mechanical strength, and other physical characteristics are different from those of the surrounding area, and examples thereof include a molten processed area, a crack area, a dielectric breakdown area, a refractive index change area, and an area where these areas are mixed. Furthermore, the mechanical strength and the like of the modified layer **220** are lower than those of the other portions of the base material **201** of the workpiece **200**.

[0058] In the modification step **1002** of the embodiment, as illustrated in FIG. **9**, the processing controller **101** of the control unit **100** of the laser beam processing machine **1** sets the focal point **211** of the laser beam **21** inside the base material **201**, and applies the laser beam **21** to the center in the width direction of the planned division lines **203** of the workpiece **200** from the back surface **208** side while controlling the laser beam application unit **20** and the moving unit **30** to relatively move the condensing unit **24** of the laser beam application unit **20** and the holding unit **10** in the X-axis direction along the planned division lines **203**. In the modification step **1002** of the embodiment, in order for the laser beam **21** to have a wavelength capable of passing through the base material **201** of the workpiece **200**, the modified layer **220** is formed inside the base material **201** along the planned division lines **203**, and the cracks **221** are formed to extend from the modified layer **220** toward the denatured portion **210** and extend from the denatured portion **210** toward the front surface **202**, as illustrated in FIG. **10**.

[0059] In this way, in the modification step **1002**, the cracks **221** that divide the device layer **205** is formed. In the modification step **1002** of the embodiment, in the laser beam processing machine **1**, the output of the laser beam **21** is set to a value that exceeds the processing threshold value of the base material **201** (in the embodiment, a value of approximately 10 times of the output in the specific layer denaturation step **1001**) in a state where the focal point **211** is set at the above-described position. Thus, the laser beam processing machine **1** forms the modified layer **220** inside the base material **201**.

[0060] In the embodiment, the laser beam **21** generated by the same oscillator **22** is used in both the specific layer denaturation step **1001** and the modification step **1002**. However, the present invention is not limited to this, and laser beams **21** generated by different oscillators **22** may be used in the specific layer denaturation step **1001** and the modification step **1002**. In the modification step **1002** of the embodiment, the laser beam processing machine **1** forms the modified layer **220** and the cracks **221** inside the workpiece **200** along all the planned division lines **203**.

[0061] In the modification step **1002** of the embodiment, the modified layer **220** and the cracks **221** are formed inside the workpiece **200** along all the planned division lines **203**, and then, the processing controller **101** of the control unit **100** of the laser beam processing machine **1** controls the moving unit **30** and the like to move the holding unit **10** to the loading/unloading area, and stops suction and holding on the holding surface **11** of the holding unit **10** in the loading/unloading area.

[0062] In the modification step **1002** of the embodiment, the processing controller **101** of the control unit **100** of the laser beam processing machine **1** controls the conveyance unit **50** to place the workpiece **200** from the holding unit **10** onto the holding surface of the spinner table **41** of the cleaning unit **40**. In the modification step **1002** of the embodiment, the processing controller **101** of the control unit **100** of the laser beam processing machine **1** sucks and holds the front surface **202** side of the workpiece **200** on the holding surface of the spinner table **41** via the tape **209** to rotate the spinner table **41** about the central axis, and drops liquid cleaning water from the cleaning nozzle **42** to the center on the back surface **208** side of the workpiece **200**.

[0063] The dropped cleaning water flows from the center side toward the outer peripheral side on the back surface **208** of the workpiece **200** due to centrifugal force generated by the rotation of the spinner table **41**, and cleans the back surface **208** of the workpiece **200**. In the modification step **1002** of the embodiment, the laser beam processing machine **1** supplies cleaning water for a predetermined period of time while rotating the spinner table **41** about the central axis, and cleans the back surface **208** of the workpiece **200**. Note that in the present invention, the workpiece **200** may not be cleaned in the modification step **1002**.

Grinding Step

[0064] Next, the grinding step **1003** will be described. FIG. **11** is a perspective view schematically illustrating the grinding step of the processing method illustrated in FIG. **3**. FIG. **12** is a cross-sectional view schematically illustrating the main parts of the workpiece after the grinding step of the processing method illustrated in FIG. **3**. The grinding step **1003** is a step of grinding and thinning the base material **201** of the workpiece **200**.

[0065] In the grinding step **1003** of the embodiment, a grinding machine **60** illustrated in FIG. **11** sucks and holds the front surface **202** side of the workpiece **200** on a holding surface **62** of a holding table **61** via the tape **209**. In the grinding step **1003** of the embodiment, the grinding machine **60** supplies grinding water while rotating a grinding wheel **64** about a central axis by a spindle **63** and rotating the holding table **61** about a central axis thereof, and causes a grinding whetstone **65** to abut on the back surface **208** of the workpiece **200** and to approach the holding table **61** at a predetermined feeding rate, thereby grinding the back surface **208** of the workpiece **200** by the grinding whetstone **65**, as illustrated in FIG. **11**.

[0066] Then, in the grinding step **1003** of the embodiment, since the modified layer **220** and the cracks **221** are formed inside the workpiece **200** as illustrated in FIG. **12**, the device layer **205** is divided into the individual devices **204** along the planned division lines **203**, with the modified layer **220** and the cracks **221** as the starting points, due to grinding stress acting from the grinding

wheel **64**. In the grinding step **1003** of the embodiment, the grinding machine **60** grinds the back surface **208** of the workpiece **200** until the workpiece **200** has a predetermined thickness.
[0067] As described above, according to the processing method and the laser beam processing program **103** according to the embodiment, the laser beam **21** is applied to the specific layer **206** included in the device layer **205** to form the denatured portion **210** by denaturing a part of the specific layer **206** in the specific layer denaturation step **1001**, before forming the modified layer **220** in the base material **201** in the modification step **1002**. According to the processing method and the laser beam processing program **103** according to the embodiment, after forming the denatured portion **210** by denaturing part of the specific layer **206** in the specific layer denaturation step **1001**, the modified layer **220** is formed in the base material **201**, and the cracks **221** extending from the modified layer **220** for dividing the device layer **205** is formed in the modification step **1002**.
[0068] As a result, according to the processing method and the laser beam processing program **103** according to the embodiment, the denatured portion **210** is formed in the specific layer **206**, whereby the cracks **221** from the modified layer **220** extend straight in the thickness direction and divide the device layer **205**. This provides the effect of reducing a division defect of the workpiece **200**.
[0069] The present invention has an effect in that it is possible to reduce a division defect.
[0070] Although the invention has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

Claims

1. A processing method of a workpiece comprising a base material and a device layer formed on the base material, the workpiece having a plurality of planned division lines, the device layer being formed by stacking a plurality of layers, the processing method comprising: denaturing a specific layer constituting the device layer by applying a laser beam along the planned division lines of the workpiece via the base material; and after denaturing the specific layer, by applying a laser beam along the planned division lines, forming a modified layer along the planned division lines in the base material and forming a crack extending from the modified layer for dividing the device layer.
2. The processing method according to claim 1, wherein denaturing the specific layer includes applying the laser beam to an area of the specific layer having a predetermined width in a direction orthogonal to an extending direction of the planned division line.
3. The processing method according to claim 2, wherein denaturing the specific layer includes applying the laser beam while positioning a focal point of the laser beam at a position different from the specific layer.
4. The processing method according to claim 2, wherein denaturing the specific layer includes applying a pulsed laser beam along the planned division lines, and the pulsed laser beam is applied such that a pulsed laser beam spot where one pulsed laser beam is applied to the specific layer and a pulsed laser beam spot where a next pulsed laser beam is applied to the specific layer partially overlap each other.
5. The processing method according to claim 1, further comprising grinding and thinning the base material after forming the modified layer and forming the crack.
6. A non-transitory computer readable storage medium storing a laser beam processing program for processing a workpiece comprising a base material and a device layer formed on the base material, the workpiece having a plurality of planned division lines, the device layer being formed by stacking a plurality of layers, the laser beam processing program causing a computer to execute: denaturing a specific layer constituting the device layer by applying a laser beam along the planned division lines of the workpiece via the base material; and after denaturing the specific layer, by

applying a laser beam along the planned division lines, forming a modified layer in the base material along the planned division lines and forming a crack extending from the modified layer for dividing the device layer.